Oussama Chatri

SECURITY-FOCUSED SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

4th-year Security of Systems engineering student at ENSTA who ships production software. Built **Keystone** (self-hosted licensing platform, six native SDKs), **ProductivityX** (offline-first Android/Desktop/Web ecosystem on one Spring Boot backend), and **WellLogAnalyzer** (hydraulics-simulation engine for petroleum engineers). Security applied end-to-end: JWT token rotation, Ed25519 offline licensing, AES-256-GCM encrypted storage, OWASP-aware development. Open to remote, hybrid, and on-site opportunities worldwide.

TECHNICAL SKILLS

Languages: Kotlin, Java, Go, Python (Basics: TypeScript, C#)

Development: Jetpack Compose (Android & Desktop), Compose Multiplatform, Android SDK, SQLDelight, JavaFX 3D, Lets-Plot

Backend: Spring Boot (Java 21), Go, REST APIs, WebSockets / STOMP, JWT auth, OpenAPI

Data & Storage: PostgreSQL, Redis, MongoDB, SQLite / SQLCipher, MinIO

Security: Application security, cryptography (Ed25519, AES-256-GCM), secure token lifecycle, device fingerprinting, OWASP-aware development

Infrastructure: Docker, GitLab CI/CD, Render, Vercel, Prometheus / OpenTelemetry

EXPERIENCE

Software Engineering Intern **Apr 2026**

Welledge / HDSS — Algiers, Algeria *On-site*

- Contributed to engineering software for industrial decision-making alongside senior engineers on production codebases
- Applied enterprise practices for architecting, building, and delivering client-facing software in a regulated industrial setting
- Applied network, code security, and commercial software licensing practices used in production systems (details limited by NDA)

PROJECTS

Keystone --- Self-Hosted Software Licensing & Entitlement Platform gitlab.com/oc-lab/keystone

- Go / Gin licensing core that issues, validates, renews, and revokes licenses across perpetual, subscription, trial, and node-locked models
- Ed25519-signed offline license packages; node-locked licenses bind to a SHA-256 device fingerprint with anti-replay leases and clock-rollback detection
- Six MIT-licensed client SDKs (Go, Kotlin, Java, JavaScript/TypeScript, Python, .NET) + two management consoles (React 19 + Compose Multiplatform desktop) over PostgreSQL 16 & Redis 7, with HMAC-signed webhooks, audit log, analytics, and Stripe billing

ProductivityX --- Offline-First Productivity Ecosystem gitlab.com/oc-lab/productivityx

- Unified notes, tasks, calendar, Pomodoro timer, and AI assistant across native Android (Kotlin / Jetpack Compose), Desktop (Compose Multiplatform), and Web (React 19), on one Spring Boot 4 backend
- Cursor-based delta sync via outbox pattern with Git-style three-way merge; encrypted local storage writes first, reconciles silently on reconnect
- JWT access/refresh token rotation bound to devices, Redis rate limiting & idempotency, STOMP WebSocket push, AI assistant streaming over SSE

WellLogAnalyzer --- Drilling Hydraulics & Well Engineering Workstation gitlab.com/oc-lab/wellloganalyzer

- Compose Multiplatform desktop app implementing SPE & API RP 13D hydraulics (ECD, annular pressure loss, bit hydraulics, surge/swab, cuttings transport) across Bingham Plastic, Power Law, and Herschel-Bulkley rheology models
- Encrypted SQLCipher vault (Argon2id key derivation), vector-clock outbox syncing to Go multi-tenant backend (MongoDB, Redis, MinIO); Lets-Plot charts, JavaFX 3D wellbore viewer, automated Word/Excel/PDF report export

Roadside Assistance Platform --- Android Client (Contributor) github.com/roadside-assistance-platform

Team project at ESI SBA. Kotlin + Jetpack Compose Android client on MVVM (Koin, Retrofit, Mapbox): request assistance, real-time provider tracking, account management.

EDUCATION

ENSTA **2025 – Present**

École Nationale Supérieure des Technologies Avancées

Engineering degree in Security of Systems — 4th year of the 5-year Engineer program

ESI SBA **2023 – 2025**

École Supérieure d'Informatique de Sidi Bel Abbés

Computer Science — first two years of the engineering program

LANGUAGES

Arabic (Native) **English** (B1) **French** (A2) **German** (A1)